

15. Show that a system can rotate while maintaining zero angular momentum.

Consider a rigid, massless plate in space. There are two masses fixed to the plate at $y = \pm 2$. Two other equal masses move in a path of $r = 1$, centered at $x = \pm 2$ on the plate. The masses start at $x = \pm 3$. The centers of rotation are fixed on the plate, and the masses stay 180° from each other about the plate center.

The right mass moves in polar coordinates centered on $x = 0, y = 2$ such that:

$$r = 1, \theta = (1 - \cos(t))\pi \text{ for } 0 \leq t \leq \pi$$

Use conservation of angular momentum to find the net rotation of the plate for one rotation of the moving masses.

$$(1) \sum m_i \vec{r}_i \times \vec{v}_i = 0$$

$$\vec{r}_P = (2 + \cos\theta) \hat{e}_r + \sin\theta \hat{e}_\phi$$

$$\vec{v}_P = (-\sin\theta \dot{\theta}) \hat{e}_r + (2 + \cos\theta) \dot{\phi} \hat{e}_\phi + \cos\theta \dot{\theta} \hat{e}_\phi - \sin\theta \dot{\phi} \hat{e}_r$$

$$\vec{r}_F = 2 \hat{e}_\phi; \vec{v}_F = -2 \dot{\phi} \hat{e}_r$$

Solve Equation 1 in MATLAB given $\vec{r}_P, \vec{v}_P, \vec{r}_F, \vec{v}_F$ using symbolic math to find expression for $\dot{\phi}$.

Numerical integration of $\dot{\phi}$ gives a change in ϕ of 0.4139 radians, or 23.7° .

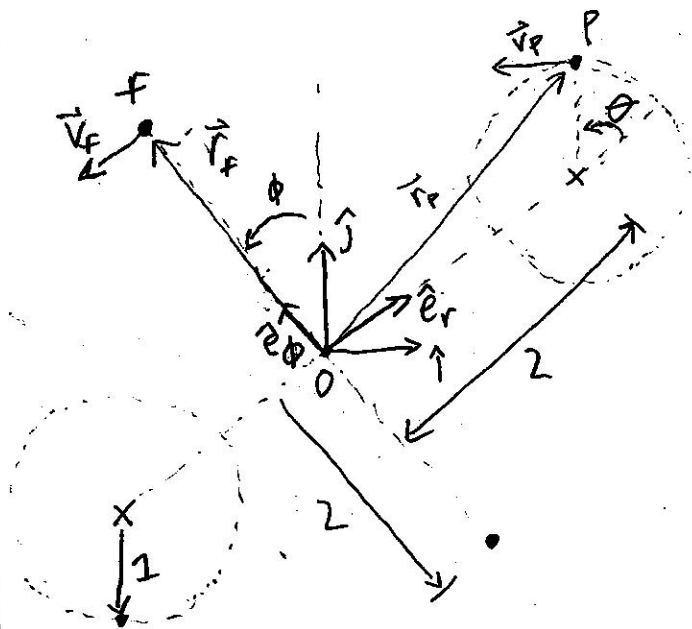


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```
function Q15_MAE57304730_F17()
```

MAE 5730: Intermediate Dynamics

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HW4: Due 9/2/2017 Problem 15 compiled as a solution for the class. Checked by Andy Ruina.

```
format compact;  
close all;  
set(0, 'DefaultAxesFontSize', 15)
```

Problem 15 - Symbolic Math and Numerical Simulation

Use conservation of angular momentum for the system to find the net rotation of the masses on the y axis at the end of one rotation of the x axis masses. Numerical integration is fine.

```
p_r = 2; % Radius to the center of the circular path of red masses  
c_r = 1; % Radius of the circular path of the red masses  
% Mass assumed to be 1 for both masses
```

```
% Because this a function file and not a script, the symbolic  
variables
```

```
% must be declared using this syntax.
```

```
th=sym('th'); thd=sym('thd'); t=sym('t'); phi=sym('phi');  
phid=sym('phid');
```

```
th = (1-cos(t))*pi; % Theta (angle of mass along circle)  
thd = sin(t)*pi; % Theta dot
```

```
% Position and velocity vectors of red mass
```

```
rp = c_r*[(p_r+cos(th)); sin(th); 0];  
vp = c_r*[(-sin(th)*thd-sin(th)*phid); ((p_r+cos(th))*phid  
+cos(th)*thd); 0];
```

```
% Position and velocity vectors of fixed mass
```

```
rf = [0; p_r; 0];  
vf = [-p_r*phid; 0; 0];
```

```
% Multiply each term by 2 for the mirrored masses
ang_mom = 2*cross(rp,vp) + 2*cross(rf,vf);

% Redundant but included for clarity
eqn = ang_mom(3) == 0;
phid_exp = solve(eqn, phid);

% Create a function handle for a symbolic expression
p.phid_func = matlabFunction(simplify(phid_exp));

init = [0];
tspan = [0 pi];

options = odeset;
options.RelTol = 1E-8; options.AbsTol = options.RelTol;

[tarray, phis] = ode45(@RHS15, tspan, init, options, p);

function zd = RHS15(t,z,p)
    zd = p.phid_func(t);
end
```

Plotting and Animation

```
% Change in plate angle
phi_change = phis(end) - phis(1)
figure;
plot(tarray, phis);
title('Change in Plate Angle Over One Cycle');
xlabel('Time (s)');
ylabel('Phi (radians)');

% Animation setup
thetas = (1-cos(tarray))*pi;

% Precompute point positions
xps = (p_r+c_r*cos(thetas)).*cos(phis) - c_r*sin(thetas).*sin(phis);
yps = (p_r+c_r*cos(thetas)).*sin(phis) + c_r*sin(thetas).*cos(phis);

xfs = -p_r*sin(phis);
yfs = p_r*cos(phis);

% For drawing the circular paths
circ_angs = linspace(0,2*pi);
circ_color = [255, 153, 153]/255;

pl_sz = 3.5; % Plate size

% Plot and save the handles for everything
fig = figure('units','normalized','outerposition',[0 0 1 1]);
hold on;
plate = patch([-pl_sz -pl_sz pl_sz pl_sz], [-pl_sz pl_sz pl_sz -
pl_sz], [0.9 0.9 0.9]);
```

```

% Red masses, center points, and paths
rp = plot([xps(1) -xps(1)], [yps(1) -
yps(1)], 'or', 'MarkerFaceColor', 'r');
cent_p = scatter([-p_r p_r], [0 0], 'rx');
circ_E = plot(p_r+c_r*cos(circ_angs),
c_r*sin(circ_angs), '--', 'Color', circ_color);
circ_W = plot(-p_r+c_r*cos(circ_angs),
c_r*sin(circ_angs), '--', 'Color', circ_color);

% Blue masses and trails
rf = plot([xfs(1) -xfs(1)], [yfs(1) -
yfs(1)], 'ob', 'MarkerFaceColor', 'b');
trail_N = plot(xfs(1), yfs(1), '--b', 'LineWidth', 1);
trail_S = plot(-xfs(1), -yfs(1), '--b', 'LineWidth', 1);

xlabel('X');
ylabel('Y');
title('Plate achieves net rotation with zero angular momentum');

axis equal;
axis([-5 5 -5 5]);
ax = gca;
ax.XAxisLocation = 'origin';
ax.YAxisLocation = 'origin';
box on

hold off;

% Animation
originalVertices = plate.Vertices;

timeFactor = 1;

num_cycles = 2; % Number of times to go back and forth
mid_pause = 0.5; % Pause at end positions

for i = 1:num_cycles
    currTime = 0;
    pause(mid_pause);
    % Reset the trail each cycle
    set(trail_N, 'XData', [], 'YData', []);
    set(trail_S, 'XData', [], 'YData', []);
    tic;

    % Go forwards
    while currTime < tspan(2)/timeFactor
        phi = interp1(tarray,phis,currTime*timeFactor);

        rotation = [cos(phi),-sin(phi); sin(phi),cos(phi)];
        plate.Vertices = (rotation*originalVertices)';

        xp = interp1(tarray,xps,currTime*timeFactor);
        yp = interp1(tarray,yps,currTime*timeFactor);

```

```

        xf = interp1(tarray, xfs, currTime*timeFactor);
        yf = interp1(tarray, yfs, currTime*timeFactor);

        rp.XData = [xp -xp]; rp.YData = [yp -yp];

        cent_x = p_r*cos(phi);
        cent_y = p_r*sin(phi);

        circ_E.XData = cent_x + c_r*cos(circ_angs); circ_E.YData =
cent_y + c_r*sin(circ_angs);
        circ_W.XData = -cent_x + c_r*cos(circ_angs); circ_W.YData = -
cent_y + c_r*sin(circ_angs);

        set(trail_N, 'XData', [trail_N.XData xf], 'YData',
[trail_N.YData yf]);
        set(trail_S, 'XData', [trail_S.XData -xf], 'YData',
[trail_S.YData -yf]);

        rf.XData = [xf -xf];
        rf.YData = [yf -yf];

        cent_p.XData = [-cent_x cent_x]; cent_p.YData = [-cent_y
cent_y];

        drawnow;
        currTime = toc;
    end
    midTime = currTime;
    pause(mid_pause);
    tic;

    % Go backwards
    while currTime > 0
        phi = interp1(tarray, phis, currTime*timeFactor);

        rotation = [cos(phi), -sin(phi); sin(phi), cos(phi)];
        plate.Vertices = (rotation*originalVertices)';

        xp = interp1(tarray, xps, currTime*timeFactor);
        yp = interp1(tarray, yps, currTime*timeFactor);
        xf = interp1(tarray, xfs, currTime*timeFactor);
        yf = interp1(tarray, yfs, currTime*timeFactor);

        rp.XData = [xp -xp]; rp.YData = [yp -yp];

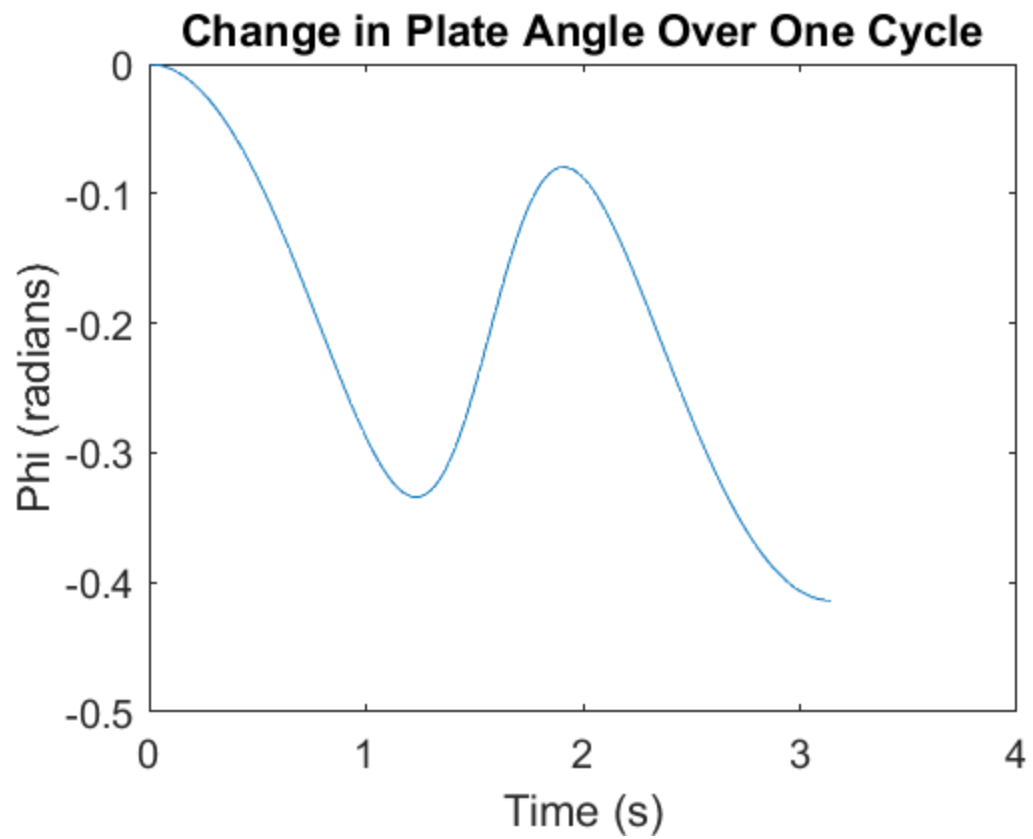
        cent_x = p_r*cos(phi);
        cent_y = p_r*sin(phi);

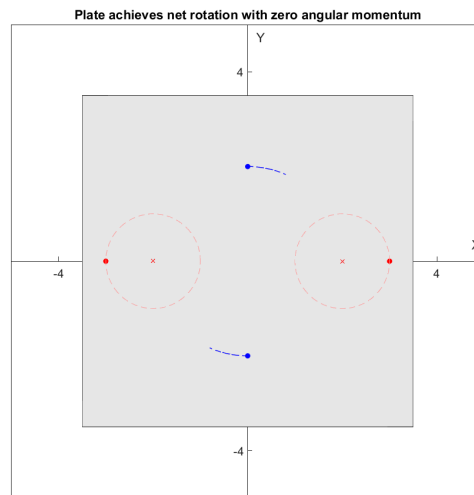
        circ_E.XData = cent_x + c_r*cos(circ_angs); circ_E.YData =
cent_y + c_r*sin(circ_angs);
        circ_W.XData = -cent_x + c_r*cos(circ_angs); circ_W.YData = -
cent_y + c_r*sin(circ_angs);

        rf.XData = [xf -xf];

```

```
        rf.YData = [yf -yf];  
  
        cent_p.XData = [-cent_x cent_x]; cent_p.YData = [-cent_y  
cent_y];  
  
        drawnow;  
        currTime = midTime - toc;  
    end  
end  
  
phi_change =  
    -0.4139
```





end

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